

KATOOMBA, AustraliaWMO: **947440**Lat: **33.712S**Lon: **150.309E**Elev: **3330**StdP: **13.01**Time Zone: **10.00 (AUS)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.0	33.6	21.0	17.8	45.6	24.4	20.9	44.9	20.8	37.1	20.2	38.5	4.1	270	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	84.6	63.4	81.0	62.5	77.4	61.6	67.6	76.9	65.9	75.0	64.3	72.6	5.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	104.0	70.8	63.1	98.0	68.4	61.8	93.3	66.9	34.1	76.9	32.6	75.3	31.3	72.4	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	16.1	12.3	29.2	90.6	1.1	4.0	28.4	93.5	27.7	95.9	27.1	98.1	26.2	101.1		
(5)			27.4	71.4	0.8	1.8	26.8	72.7	26.4	73.8	25.9	74.8	25.3	76.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	63.5	63.3	59.6	54.9	49.2	44.6	42.6	44.5	50.1	53.9	57.1	61.6
	DBStd	9.21	6.12	6.48	5.54	5.20	4.59	4.30	3.51	4.84	6.03	6.29	8.01	6.51
	HDD50	794	0	1	1	15	70	169	231	184	72	30	19	2
	HDD65	4329	102	101	185	304	491	613	696	636	446	347	262	146
	CDD50	2136	417	373	299	163	43	6	1	13	76	151	233	361
	CDD65	196	55	53	18	1	0	0	0	0	0	3	26	40
	CDH74	1452	439	342	112	6	0	0	0	0	0	22	198	333
	CDH80	362	128	79	18	0	0	0	0	0	0	1	59	77
(14) Wind	WSAvg	3.8	3.1	2.8	2.8	2.6	3.9	4.9	4.7	5.4	4.9	4.2	3.6	3.2
(15) Precipitation	PrecAvg	54.7	6.5	6.9	7.0	4.6	4.3	3.9	2.4	3.5	2.7	4.1	4.7	4.6
	PrecMax	84.8	19.4	21.5	23.7	23.5	12.6	17.5	10.5	22.9	8.3	16.5	16.9	12.8
	PrecMin	29.5	0.6	0.8	0.5	0.2	0.2	0.0	0.0	0.0	0.2	0.3	0.4	0.1
	PrecStd	15.4	4.3	4.7	5.0	4.7	3.5	4.0	2.2	4.8	2.0	3.6	3.6	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	87.5	82.6	74.3	64.4	59.2	55.7	65.4	70.8	77.4	86.5	86.1
		MCWB	65.7	65.4	63.6	58.1	54.5	49.9	46.1	52.1	55.1	55.8	60.0	63.0
	2%	DB	84.3	82.4	76.8	69.7	61.0	56.3	52.6	59.8	66.9	72.9	80.0	82.3
		MCWB	63.7	65.2	62.2	57.1	52.3	49.2	45.1	48.8	52.6	55.4	59.9	61.6
	5%	DB	79.4	78.5	72.3	66.5	58.2	53.9	50.3	56.6	63.7	69.0	74.2	78.4
		MCWB	62.0	64.7	60.1	55.9	51.3	48.3	44.6	47.0	51.2	54.1	58.5	60.7
	10%	DB	75.4	74.5	68.7	63.5	55.9	51.8	48.4	53.5	60.2	65.4	70.5	74.5
		MCWB	61.4	63.3	59.3	55.1	50.7	47.2	43.5	45.8	50.1	52.5	57.6	60.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	71.1	68.2	61.4	57.5	53.6	49.4	53.6	57.3	61.3	64.0	67.7
		MCDB	82.3	78.1	74.6	68.1	59.7	56.8	51.3	59.8	67.2	70.8	76.1	77.5
	2%	WB	67.1	68.1	64.8	59.2	55.3	51.3	47.5	50.6	55.0	58.1	62.4	65.4
		MCDB	78.0	76.2	71.5	65.3	58.1	54.2	50.0	57.1	63.0	66.5	73.0	75.2
	5%	WB	65.2	66.2	62.7	57.6	53.6	49.6	46.1	48.6	53.4	56.4	60.9	63.8
		MCDB	74.4	75.0	69.0	63.3	56.1	52.5	48.4	53.8	59.5	64.5	71.5	72.5
	10%	WB	63.5	64.5	60.8	56.1	51.9	47.9	44.9	46.6	51.6	54.6	59.0	62.0
		MCDB	71.1	72.8	66.4	61.5	54.7	50.8	47.2	52.5	57.8	62.1	66.7	70.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	14.8	14.3	13.5	10.8	10.0	10.7	13.3	14.7	16.4	15.6	17.9
		MCDBR	24.3	21.9	20.6	18.6	14.5	13.3	13.4	17.3	18.1	21.3	22.3	23.7
		MCWBR	10.1	9.9	9.9	9.5	8.3	8.3	8.3	9.0	8.9	9.8	9.9	9.8
	5% WB	MCDBR	19.4	18.3	17.3	15.0	10.1	10.7	9.9	14.8	15.3	18.0	18.9	19.9
		MCWBR	10.1	9.7	9.6	8.7	6.5	7.7	7.0	8.8	8.5	10.2	9.9	10.4
(40) Clear-Sky Solar Irradiance	taub		0.338	0.334	0.313	0.292	0.278	0.267	0.262	0.266	0.292	0.311	0.330	0.330
	taud		2.525	2.562	2.597	2.635	2.624	2.672	2.677	2.665	2.568	2.533	2.516	2.534
	Ebn at Noon		317	312	307	298	286	283	290	304	310	316	318	321
	Edh at Noon		35	33	30	25	23	20	21	24	30	33	35	35
(44) All-Sky Solar Radiation	RadAvg		2072	1759	1507	1228	976	764	891	1176	1575	1832	1936	2131
	RadStd		162	171	142	109	67	66	64	78	103	190	187	204

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	+0.81	N/A	N/A	+1.37	N/A	N/A	-4	-155	+333	+178

Nomenclature: See separate page